

MASTER_CONTEXT.md

Part 6 — Database & API Blueprint

Document Purpose

This section provides a consolidated blueprint of the database architecture and API ecosystem that powers Aetheris Trade.

Every backend implementation must follow this blueprint.

Backend Philosophy

The backend must be:

- Secure
 - Modular
 - Scalable
 - Performant
 - Maintainable
 - AI Ready
-

System Flow

User

↓

Frontend (Next.js)

↓

API Layer

↓

Business Logic

↓

Database

↓

Response

Core Database Modules

Users

Portfolio

Holdings

Transactions

Stocks

Market Data

Watchlist

Learning

Competitions

Leaderboard

Notifications

AI Conversations

Entity Relationships

User

|

├─ Portfolio

|

├─ Holdings

|

├─ Transactions

|

└─ Analytics

|

├─ Watchlist

|

├─ AI Conversations

|

└─ Learning Progress
|
└─ Notifications
|
└─ Competition Entries

Stock
|
└─ Market Data
└─ Historical Prices
└─ Holdings
└─ Transactions
└─ Watchlist

Database Principles

- Normalized Schema
- Strong Relationships
- Indexed Queries
- Transaction Safety
- No Duplicate Data
- Future Scalability

Authentication

Provider

- Clerk

Flow

User

↓

Clerk

↓

JWT

↓

Protected API

↓

Database

User Roles

- Guest
- User
- Admin

API Structure

Base URL

/api/v1

API Modules

Authentication

Users

Portfolio

Holdings

Transactions

Stocks

Market

Watchlist

Learning

AI

Competitions

Leaderboard

Notifications

Admin

Request Lifecycle

Request

↓

Authentication

↓

Validation

↓

Controller

↓

Service

↓

Repository

↓

Database

↓

Response

API Standards

Every API must be:

- RESTful
- Stateless
- Versioned
- JSON Only

Standard Response

Success

```
{
  "success": true,
  "data": {},
  "message": ""
}
```

Error

```
{
  "success": false,
  "error": {
    "code": "",
    "message": ""
  }
}
```

Trading Flow

Buy Request

↓

Validation

↓

Portfolio Check

↓

Transaction

↓

Holding Update

↓

Analytics Update

↓

Success Response

AI Flow

User Prompt

↓

Prompt Builder

↓

Context Engine

↓

LLM

↓

Response Validation

↓

User

Learning Flow

Course

↓

Lesson

↓

Quiz

↓

Result

↓

Progress

↓

Certificate

Competition Flow

Competition

↓

Registration

↓

Trading

↓

Ranking

↓

Leaderboard

↓

Rewards

Security

Mandatory

- JWT Validation
- Role Verification
- Input Validation
- Output Sanitization
- HTTPS

- Rate Limiting
 - Audit Logs
-

Performance Targets

API

<500ms

Trading

<700ms

AI

<5 Seconds

Database Performance

- Indexed Queries
 - Optimized Relations
 - Pagination
 - Query Optimization
 - Read Scalability
-

Future Scalability

Support

- 1M+ Users
 - Read Replicas
 - Redis Cache
 - CDN
 - Queue System
 - Background Workers
-

AI Backend Rules

AI services must:

- Never bypass authentication
 - Never access unauthorized data
 - Always validate context
 - Log AI interactions
 - Respect user privacy
-

End of Part 6

Next:

- Part 7 — Development Workflow